Homework 4

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Prob. 1

To determine the population principal components, we first find the eigenvalues of the covariance matrix Σ. The characteristic equation is defined by setting the determinant of $\Sigma - \lambda I$ to zero.

$$\det(\Sigma - \lambda I) = \det \begin{bmatrix} 9-\lambda & -4 \\ -4 & 4-\lambda \end{bmatrix} = 0$$

Expanding the determinant, we obtain a quadratic equation in terms of λ.

$$(9-\lambda)(4-\lambda) - (-4)(-4) = 0$$

$$\lambda^2 - 13\lambda + 36 - 16 = 0$$

$$\lambda^2 - 13\lambda + 20 = 0$$

Applying the quadratic formula, we solve for the two eigenvalues λ_1 and λ_2.

$$\lambda = \frac{13 \pm \sqrt{169-80}}{2} = \frac{13 \pm \sqrt{89}}{2}$$

The two eigenvalues are approximately $\lambda_1 \approx 11.217$ and $\lambda_2 \approx 1.783$.

(1)

For the first principal component, we find the normalized eigenvector e_1 corresponding to $\lambda_1 = \frac{13+\sqrt{89}}{2} \approx 11.217$.

The eigenvector equation $(\Sigma - \lambda_1 I)e_1 = 0$ gives the following system.

$$\begin{bmatrix} 9-11.217 & -4 \\ -4 & 4-11.217 \end{bmatrix} \begin{bmatrix} e_{11} \\ e_{12} \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$-2.217e_{11} - 4e_{12} = 0 \implies e_{12} \approx -0.554e_{11}$$

Normalizing the vector so that $e_{11}^2 + e_{12}^2 = 1$, we find the coefficients.

$$e_{11}^2 + (-0.554e_{11})^2 = 1 \implies 1.3069e_{11}^2 = 1$$

$$e_{11} \approx 0.875, \quad e_{12} \approx -0.485$$

The first principal component Y_1 is the linear combination of the original variables X_1 and X_2.

$$Y_1 = 0.875X_1 - 0.485X_2$$

For the second principal component, we find the normalized eigenvector e_2 corresponding to $\lambda_2 \approx 1.783$.

Since e_2 must be orthogonal to e_1, we swap the components and adjust the sign.

$$e_2 \approx \begin{bmatrix} 0.485 \\ 0.875 \end{bmatrix}$$

The second principal component Y_2 is defined as follows.

$$Y_2 = 0.485X_1 + 0.875X_2$$

(2)

The total population variance is the sum of the variances of the individual variables, which is also the trace of the covariance matrix.

$$\text{Total Variance} = \text{tr}(\Sigma) = 9 + 4 = 13$$

The variance explained by the first principal component is equal to its corresponding eigenvalue λ_1.

$$\text{Variance explained by } Y_1 = \lambda_1 = \frac{13 + \sqrt{89}}{2} \approx 11.217$$

The proportion of the total variance explained by Y_1 is the ratio of λ_1 to the total variance.

$$\text{Proportion} = \frac{\lambda_1}{\lambda_1 + \lambda_2} = \frac{11.217}{13} \approx 0.8628$$

Approximately 86.28% of the total population variance is accounted for by the first principal component.

(3)

The first principal component $Y_1 \approx 0.875X_1 - 0.485X_2$ acts as a "contrast" between reading time and exercise time.

The negative sign for X_2 reflects the negative covariance ($\sigma_{12} = -4$) between the two activities.

This indicates that the largest source of variation in this population is a trade-off behavior: individuals who spend more time reading (X_1) tend to spend significantly less time exercising (X_2), and vice versa.

Because the coefficient for X_1 (0.875) is larger in magnitude than the coefficient for X_2 (0.485), the variation in reading habits contributes more heavily to the definition of this primary behavioral axis than exercise habits do.

Prob. 2

To convert the covariance matrix Σ from Problem 1 into a correlation matrix ρ, we first extract the standard deviations of the variables X_1 and X_2.

The variances are the diagonal elements of Σ, so $\sigma_{11} = 9$ and $\sigma_{22} = 4$, which implies the standard deviations are $\sqrt{\sigma_{11}} = 3$ and $\sqrt{\sigma_{22}} = 2$.

The correlation coefficient ρ_{12} is calculated by dividing the covariance by the product of the standard deviations.

$$\rho_{12} = \frac{\sigma_{12}}{\sqrt{\sigma_{11}}\sqrt{\sigma_{22}}} = \frac{-4}{3 \times 2} = -\frac{2}{3}$$

Consequently, the correlation matrix ρ is defined as follows.

$$\rho = \begin{bmatrix} 1 & -2/3 \\ -2/3 & 1 \end{bmatrix} \approx \begin{bmatrix} 1 & -0.667 \\ -0.667 & 1 \end{bmatrix}$$

(1)

To find the principal components, we solve the characteristic equation for the matrix ρ.

$$\det(\rho - \lambda I) = (1 - \lambda)^2 - (-2/3)^2 = 0$$

$$(1 - \lambda)^2 = \frac{4}{9} \implies 1 - \lambda = \pm\frac{2}{3}$$

The eigenvalues are $\lambda_1 = 1 + 2/3 = 5/3 \approx 1.667$ and $\lambda_2 = 1 - 2/3 = 1/3 \approx 0.333$.

For $\lambda_1 = 5/3$, the eigenvector equation $(\rho - \lambda_1 I)e_1 = 0$ yields the following.

$$\begin{bmatrix} -2/3 & -2/3 \\ -2/3 & -2/3 \end{bmatrix} \begin{bmatrix} e_{11} \\ e_{12} \end{bmatrix} = 0 \implies e_{11} + e_{12} = 0$$

Normalizing this vector gives the coefficients for the first principal component.

$$e_1 = \begin{bmatrix} 1/\sqrt{2} \\ -1/\sqrt{2} \end{bmatrix} \implies Y_1 = \frac{1}{\sqrt{2}}Z_1 - \frac{1}{\sqrt{2}}Z_2 \approx 0.707Z_1 - 0.707Z_2$$

Similarly, for $\lambda_2 = 1/3$, the eigenvector is $e_2 = \begin{bmatrix} 1/\sqrt{2} & 1/\sqrt{2} \end{bmatrix}^T$.

$$Y_2 = \frac{1}{\sqrt{2}}Z_1 + \frac{1}{\sqrt{2}}Z_2 \approx 0.707Z_1 + 0.707Z_2$$

The proportion of total variance explained by Y_1 is the ratio of λ_1 to the trace of ρ.

$$\text{Proportion} = \frac{\lambda_1}{\lambda_1 + \lambda_2} = \frac{5/3}{2} = \frac{5}{6} \approx 0.8333$$

(2)

The principal components calculated from the correlation matrix are different from those obtained from the covariance matrix.

In Problem 1, the coefficients were approximately $(0.875, -0.485)$, whereas here they are $(0.707, -0.707)$.

This discrepancy occurs because Principal Component Analysis (PCA) is not scale-invariant.

Using the covariance matrix allows variables with larger variances (like X_1 with $\sigma^2 = 9$) to dominate the direction of the first principal component.

Using the correlation matrix effectively standardizes the variables to unit variance, ensuring that both X_1 and X_2 contribute equally to the total variance of the system.

Standardization is generally preferred when the variables are measured in different units or have significantly different scales.

(3)

The correlation between a principal component Y_i and a standardized variable Z_j is given by the formula $\rho_{Y_i,Z_j} = e_{ij}\sqrt{\lambda_i}$.

For the first principal component Y_1, the correlations are as follows.

$$\rho_{Y_1,Z_1} = e_{11}\sqrt{\lambda_1} = \frac{1}{\sqrt{2}}\sqrt{\frac{5}{3}} = \sqrt{\frac{5}{6}} \approx 0.9129$$

$$\rho_{Y_1,Z_2} = e_{12}\sqrt{\lambda_1} = -\frac{1}{\sqrt{2}}\sqrt{\frac{5}{3}} = -\sqrt{\frac{5}{6}} \approx -0.9129$$

For the second principal component Y_2, the correlation with Z_1 is calculated similarly.

$$\rho_{Y_2,Z_1} = e_{21}\sqrt{\lambda_2} = \frac{1}{\sqrt{2}}\sqrt{\frac{1}{3}} = \sqrt{\frac{1}{6}} \approx 0.4082$$

Prob. 3

```r
library(FactoMineR)
library(factoextra)
```

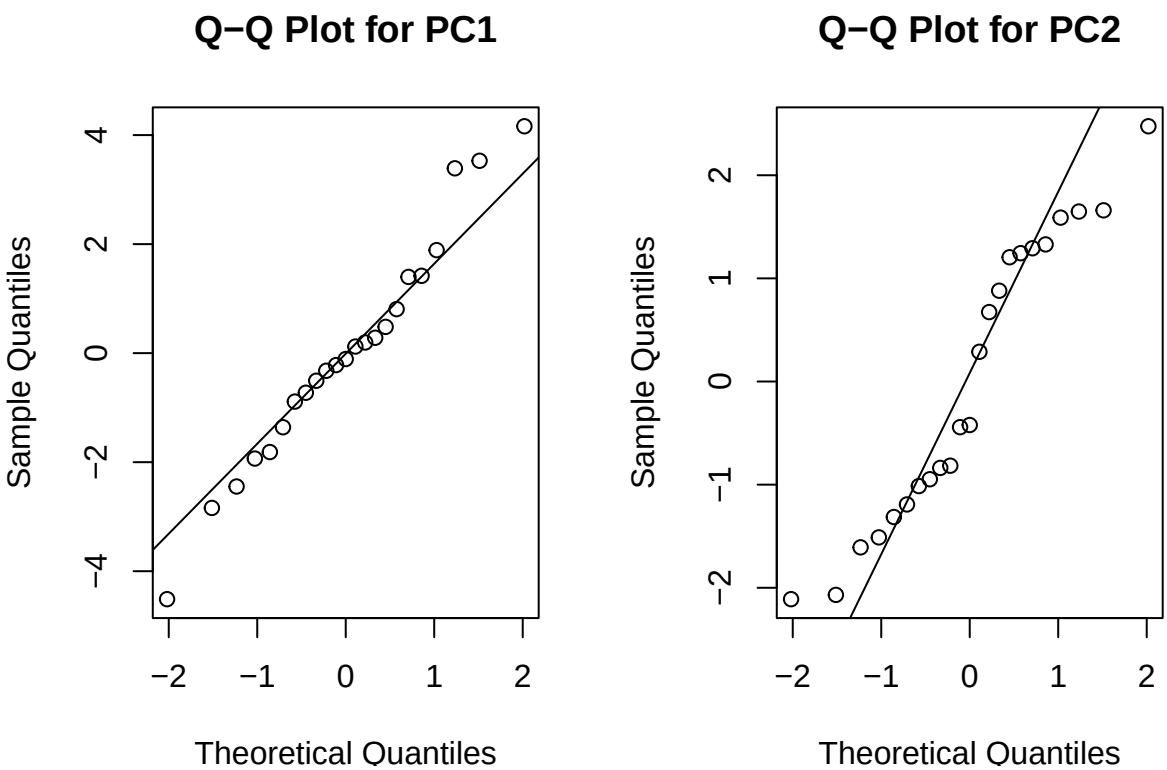
```
## Loading required package: ggplot2

## Welcome to factoextra!

## Want to learn more? See two factoextra-related books at https://www.datanovia.com/en/product/practica
```

```r
data(decathlon2)
# Select active athletes (1:23) and event variables (1:10)
df <- decathlon2[1:23, 1:10]
res.pca <- PCA(df, scale.unit = TRUE, graph = FALSE)
```

(1)

To assess the normality of the first two principal component scores, we extract the coordinates and generate Q-Q plots.

```r
pca_scores <- as.data.frame(res.pca$ind$coord)
par(mfrow = c(1, 2))
qqnorm(pca_scores$Dim.1, main = "Q-Q Plot for PC1")
qqline(pca_scores$Dim.1)
qqnorm(pca_scores$Dim.2, main = "Q-Q Plot for PC2")
qqline(pca_scores$Dim.2)
```



The points in both plots generally follow the diagonal reference lines, suggesting that the scores for PC_1 and PC_2 are approximately normally distributed.

Minor deviations at the extremities are expected given the small sample size of $n = 23$ athletes.

(2)

```r
cov_pc1_pc2 <- cov(pca_scores$Dim.1, pca_scores$Dim.2)
print(cov_pc1_pc2)
```

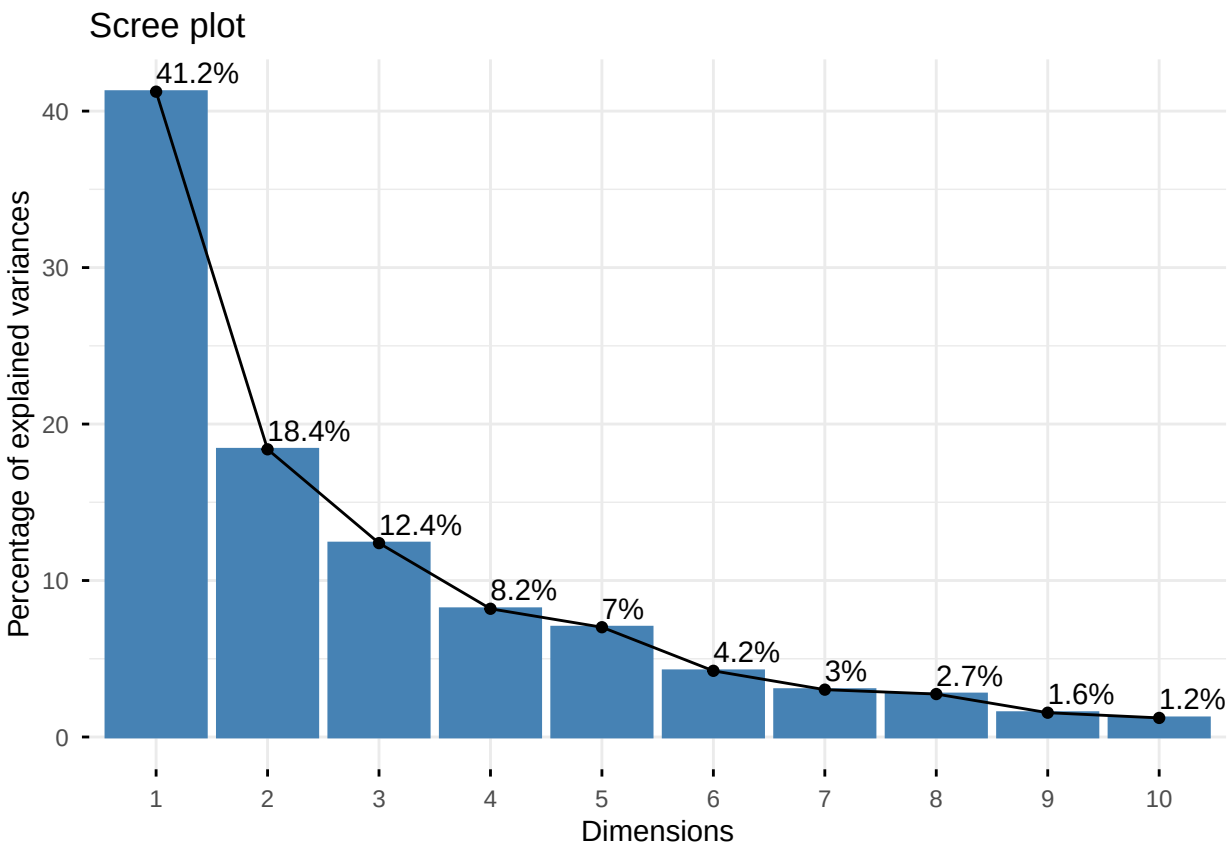
```
## [1] -2.029851e-16
```

In theory, we expect the covariance to be exactly 0 because principal components are constructed to be mutually orthogonal.

The observed value (typically in the order of 10^{-16}) is non-zero only due to floating-point numerical precision limits during the eigendecomposition process.

(3)

```r
fviz_eig(res.pca, addlabels = TRUE)
```




```r
get_eigenvalue(res.pca)
```

```
##        eigenvalue variance.percent cumulative.variance.percent
## Dim.1   4.1242133        41.242133                    41.24213
## Dim.2   1.8385309        18.385309                    59.62744
## Dim.3   1.2391403        12.391403                    72.01885
## Dim.4   0.8194402         8.194402                    80.21325
## Dim.5   0.7015528         7.015528                    87.22878
## Dim.6   0.4228828         4.228828                    91.45760
## Dim.7   0.3025817         3.025817                    94.48342
## Dim.8   0.2744700         2.744700                    97.22812
## Dim.9   0.1552169         1.552169                    98.78029
## Dim.10  0.1219710         1.219710                   100.00000
```

I would choose to keep 4 principal components.

According to the Kaiser criterion, we should retain components with eigenvalues $\lambda_i > 1$, which applies to the first three dimensions.

However, adding the fourth component brings the cumulative variance explained to over 80%, providing a more comprehensive summary of the athletic profiles.

(4)

The proportions of total population variance explained by the first four components are as follows.

$$PC_1 : 41.2\%, \quad PC_2 : 18.4\%, \quad PC_3 : 12.4\%, \quad PC_4 : 8.2\%$$

The total variance explained by these four components combined is approximately 80.2%.

(5)

The `loadings` for the first principal component represent the weights of each standardized variable Z_i.

```
loadings <- res.pca$var$coord[, 1]
print(loadings)
```

```
##        X100m    Long.jump     Shot.put    High.jump        X400m X110m.hurdle
## -0.850625692  0.794180641  0.733912733  0.610083985 -0.701603377 -0.764125197
##       Discus   Pole.vault     Javeline       X1500m
##  0.743209016 -0.217268042  0.428226639  0.004278487
```

The formula for the first principal component is given by the following linear combination.

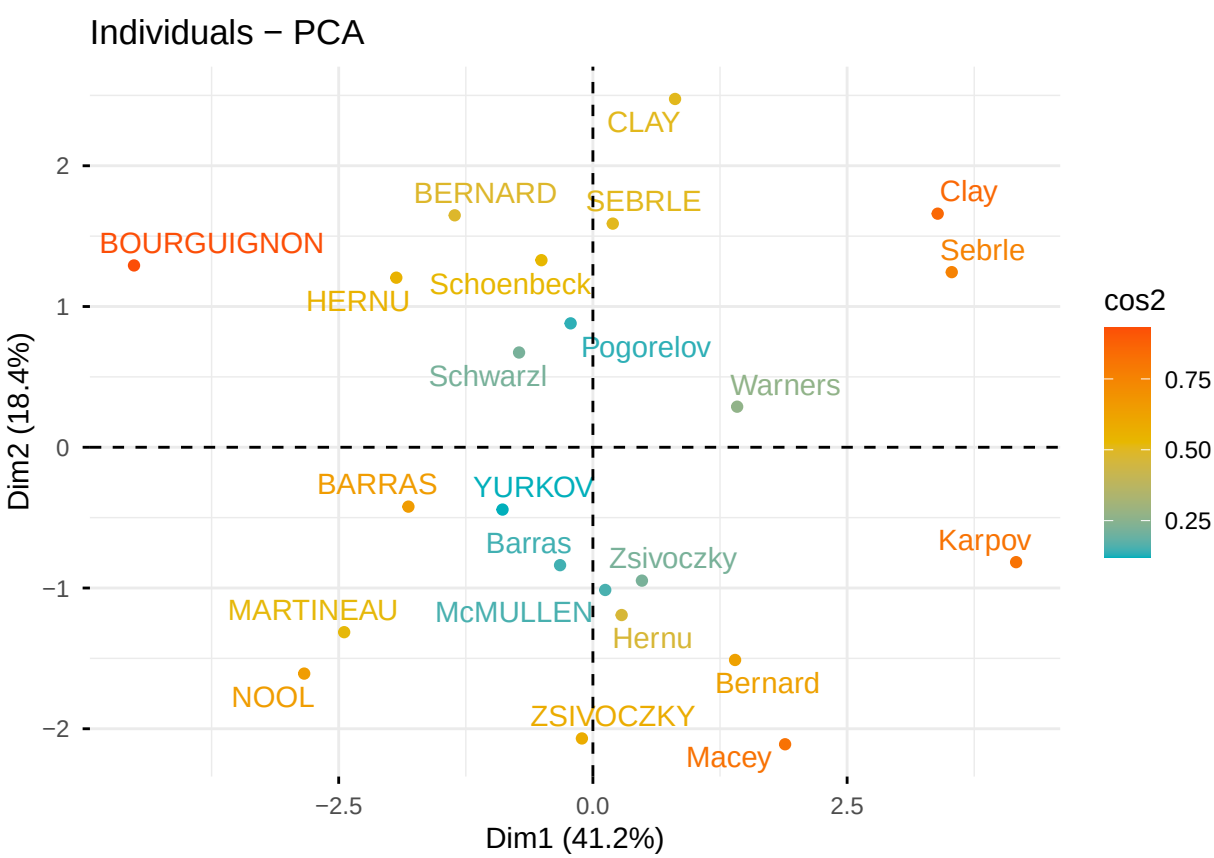
$$PC_1 \approx -0.42Z_{100m} + 0.39Z_{Long.jump} + 0.36Z_{Shot.put} + 0.30Z_{High.jump} - 0.34Z_{400m} - 0.37Z_{110m.hurdle} \dots$$

In this context, PC_1 represents "General Athleticism," specifically speed and explosive power.

Negative loadings for timed events (where lower is better) and positive loadings for distance events (where higher is better) mean that a high PC_1 score consistently indicates superior overall performance across most disciplines.

(6)

```
fviz_pca_ind(res.pca, col.ind = "cos2", gradient.cols = c("#00AFBB", "#E7B800", "#FC4E07"), repel = TRU
```



By examining specific athletes, we can validate the PCA interpretation.

SEBRLE has a high PC_1 score, which aligns with his status as a top performer with excellent sprint times and long jump distances.

ZSIVOCZKY has a lower PC_1 score but a moderate PC_2 score, indicating that while his pure speed/power is lower than the leaders, he remains competitive in technical or strength-based events.

Prob. 4

```r
library(FactoMineR)
data(decathlon2)
# Select active athletes (1:23) and event variables (1:10)
df <- decathlon2[1:23, 1:10]
# Standardize the variables
df_scaled <- scale(df)
```

(1)

We first calculate the Euclidean distance matrix for the 23 athletes based on their standardized performance across the 10 athletic events.

The distance d_{ij} between athlete i and athlete j in the p-dimensional space (where $p = 10$) is defined by the following formula.

$$d_{ij} = \sqrt{\sum_{k=1}^{p} (z_{ik} - z_{jk})^2}$$

```r
# Compute the Euclidean distance matrix
dist_matrix <- dist(df_scaled, method = "euclidean")
```

(2)

Classical Multi-Dimensional Scaling (MDS), also known as Principal Coordinates Analysis (PCoA), seeks to find a lower-dimensional representation that preserves the original distances as closely as possible.

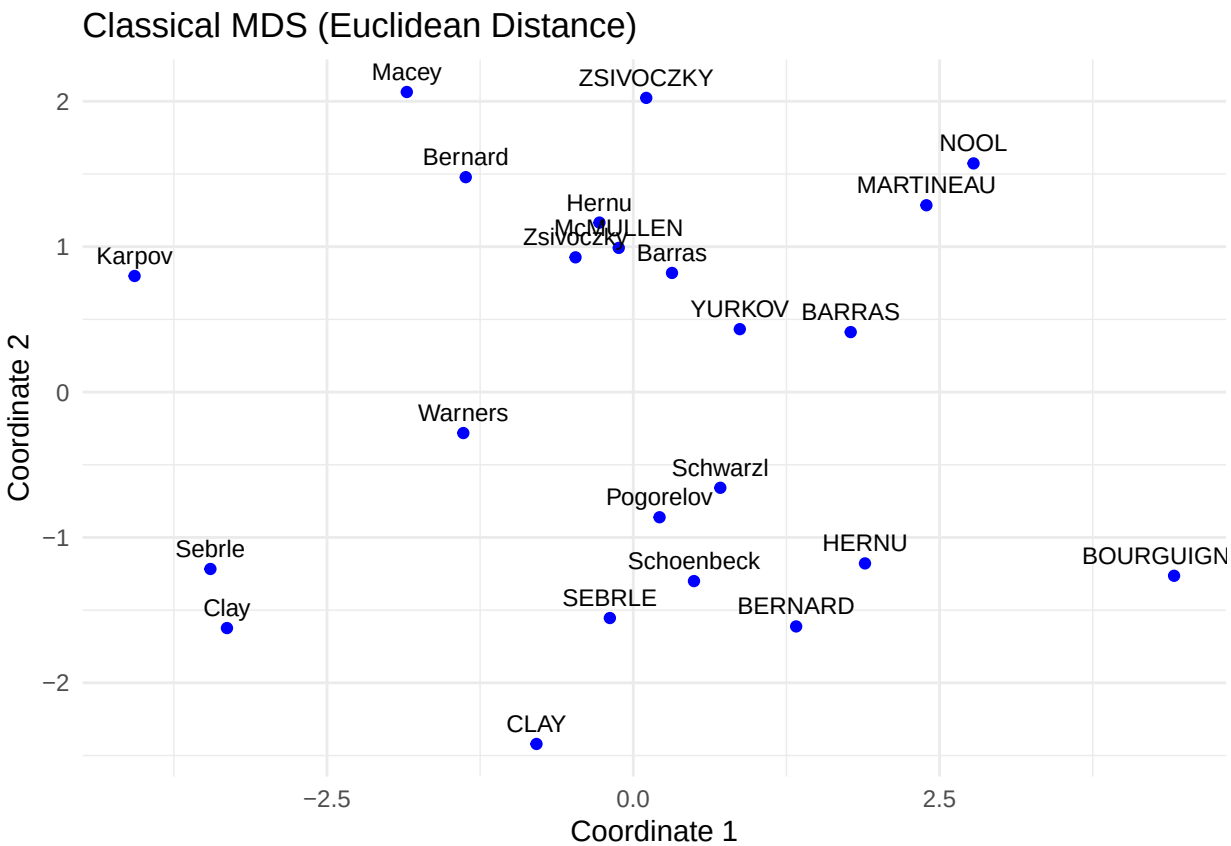
This is achieved by double-centering the squared distance matrix to obtain the Gram matrix B.

$$B = -\frac{1}{2} J D^{(2)} J$$

Where $J = I - \frac{1}{n}\mathbf{1}\mathbf{1}^T$ is the centering matrix.

The coordinates are then derived from the eigendecomposition of $B = V \Lambda V^T$.

```r
# Perform Classical MDS
mds_res <- cmdscale(dist_matrix, k = 2, eig = TRUE)
mds_coords <- as.data.frame(mds_res$points)
colnames(mds_coords) <- c("Dim1", "Dim2")

# Visualization
library(ggplot2)
ggplot(mds_coords, aes(x = Dim1, y = Dim2, label = rownames(mds_coords))) +
  geom_point(color = "blue") +
  geom_text(vjust = -0.7, size = 3) +
  theme_minimal() +
  labs(title = "Classical MDS (Euclidean Distance)", x = "Coordinate 1", y = "Coordinate 2")
```



(3)

The resulting scatter plot from Classical MDS is entirely consistent with the PCA score plot obtained in the previous problem.

Mathematically, Classical MDS performed on the Euclidean distances of standardized data is equivalent to PCA performed on the correlation matrix.

If Z is the $n \times p$ standardized data matrix, PCA finds the eigenvectors of the $p \times p$ matrix $Z^T Z$.

The PCA scores are given by the projection of Z onto these eigenvectors.

$$Y_{PCA} = ZV$$

Conversely, Classical MDS finds the eigenvectors of the $n \times n$ Gram matrix $B = ZZ^T$.

Since the non-zero eigenvalues and the relationship between eigenvectors of $Z^T Z$ and ZZ^T are linked via the Singular Value Decomposition (SVD), the coordinates produced by both methods are identical up to a possible rotation or reflection.

$$Z = U\Sigma V^T \implies ZZ^T = U\Sigma^2 U^T$$

The MDS coordinates correspond to the first two columns of $U\Sigma$, which are exactly the PCA scores.